BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 6471 / 6478

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Paper: I Objective (i)

فزکس

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6471

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جوابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. In 8.70×10^4 kg has number of significant digits:

(A) Four (B) Two

(C) Three (D) Seven

Q2. If percentage uncertainty in radius of a sphere is 2%, then total uncertainty in the volume is:

(A) 6% (B) 2%

(C) 4% (D) 8%

Q3. A single vector having same effect as all the original vectors taken together is:

(A) Equal vector (B) Resultant vector

(C) Position vector (D) Unit vector

Q4. Which is vector quantity?

(A) Length
(B) Volume
(C) Work
(D) Velocity

Q5. The distance covered by a body in time 't' starting from rest is:

(A) at^2
(B) v^2t
(C) $a^2t/2$
(D) $at^2/2$

Q6. The slope of velocity-time graph gives:

(A) Speed
(B) Torque
(C) Displacement
(D) Acceleration

Q7. Earth receives large amount of energy directly from:

(A) Wind
(B) Sun
(C) Moon
(D) Water

Q8. A rev min^{-1} is equal to:

(A) $\pi/30$ rad s^{-1}
(B) $\pi/60$ rad s^{-1}
(C) $\pi/15$ rad s^{-1}
(D) $\pi/20$ rad s^{-1}

Q9. Which is correct relation?

(A) $v = w \times r$
(B) $v = w.r$
(C) $w = v \times r$
(D) $v = r \times w$

Q10. The maximum drag force on falling sphere is 9.8 N, its weight is:

(A) 1 N
(B) 9.8 N
(C) 19.8 N
(D) 980 N

Q11. The wave form of SHM is:

(A) Square wave
(B) Saw tooth wave
(C) Sine wave
(D) Random wave

Q12. If the pressure of gas is doubled, then speed of sound is:

(A) Doubled
(B) Not changed
(C) Half
(D) Increased 4 times

Q13. Velocity of sound in free space is:

(A) 332 ms^{-1}
(B) 224 ms^{-1}
(C) 76 ms^{-1}
(D) Zero ms^{-1}

Q14. Thin oil film on water surface shows colour due to:

(A) Diffraction
(B) Interference
(C) Polarization
(D) Dispersion

Q15. The ratio of size of image to the size of object is called:

(A) Polarization
(B) Magnification
(C) Transmission
(D) Reflection

Q16. Entropy of an irreversible process:

(A) Increases
(B) Decreases
(C) Remains same
(D) None of these

Q17. If 32 Joule work is done by absorbing heat of 42 Joule, change in internal energy is:

(A) $\Delta U = 74J$
(B) $\Delta U = 10J$
(C) $\Delta U = 21/8$ J
(D) $\Delta U = 8/21$ J

Paper: I Subjective

فزکس

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any THREE questions from Section C.

نوٹ: حصہ دوم لازمی ہے۔ حصہ سوئم میں سے کوئی سے تین سوالات کے جوابات تحریر کیجیے۔

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) The period of simple pendulum is measured by a stop watch. What type of errors are possible in the time period?

(ii) Write down the dimensions of a) pressure b) density

(iii) What is the light year and write down its units?

(iv) Write down any two rules for rounding off the significant figures.

(v) Can you add zero to a null vector?

(vi) The vector sum of three vectors gives a zero resultant. What can be the orientation of the vectors?

(vii) What is the moment of a force about the point lying on the axis of rotation?

(viii) An object is thrown vertically upward. Discuss the sign of acceleration due to gravity, relative to velocity, while the object is in air.

(ix) What is meant by instantaneous acceleration? Write down its formula.

(x) Show that $F = \Delta P/\Delta t$

(xi) How energy can be obtained from sea tides? Explain.

(xii) When a rocket re-enters the atmosphere, its nose cone becomes very hot. Where does this heat energy come from?

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) During circular motion with constant speed, body's velocity is northward. After time Δt its velocity is westward. What will be the direction of centripetal force? Explain with vector diagram.

(ii) Explain the difference between tangential velocity and angular velocity.

(iii) What is meant by moment of inertia? Explain its significance.

(iv) Show that orbital angular momentum $Lo=mvr$

(v) Why fog droplets appear to be suspended in air?

(vi) What is the difference between laminar flow and turbulent flow?

(vii) Find the angular velocity of a point on the earth's equator as a result of earth's rotation.

(viii) A simple pendulum has time period 2s with amplitude 0.10m. Find its maximum acceleration.

(ix) Name two characteristics of simple harmonic motion.

(x) Does frequency depend on amplitude for harmonic oscillators?

(xi) State law of reflection for transverse waves.

(xii) What features do longitudinal waves have in common with transverse waves?

Q4. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) How would you manage to get more orders of spectra using a diffraction grating?

(ii) What do sugar and tartaric acid show when they are in solution and why?

(iii) How did Michelson measure the length of a standard metre?

(iv) What will be the angular magnification of telescope having an objective of focal length of 20cm and an eye piece of 4.0cm, both convex lenses.

(v) Describe the function of repeaters and photodiode in signal transmission?

(vi) A magnifying glass gives a five-times enlarged image at a distance of 25 cm from the lens. Find focal length using ray diagram.

(vii) Why do we keep pressure or volume constant to study the effect of heating gases?

(viii) A thermo flask containing milk as a system is shaken rapidly. Does the temperature of milk rise?

(ix) Why the internal energy is similar to gravitational potential energy? Explain

> Note: Attempt any THREE questions. Each question carries Eight (8) marks. (8x3=24)

Q5. (Marks 8)

(a) [5] Discuss vector addition of a number of coplanar vectors by rectangular components.

(b) [3] A bomber dropped a bomb at a height of 490m when its velocity along horizontal was 300kmh^{-1}. How long was it in air?

Q6. (Marks 8)

(a) [5] Derive an expression of Laplace's correction for speed of sound in air. Is this correction close to experimental value?

(b) [3] A car of mass 800kg travelling at 54kmh^{-1} is brought to rest in 60 metres. Find the average retarding force on the car.

Q7. (Marks 8)

(a) [5] Define and explain centripetal force and derive the relation for it.

(b) [3] A block of mass 4.0kg is dropped from a height of 0.80m onto a spring of spring constant K=1960 Nm^{-1}. Find the maximum distance through which the spring will be compressed.

Q8. (Marks 8)

(a) [5] Define and explain the velocity at which the water droplets attain the dynamic equilibrium in air.

(b) [3] Calculate the entropy change when 1.0kg ice at 0° C melts into water at 0° C. Latent heat of fusion of ice Lf=3.36x10^{5} J kg^{-1}.

Q9. (Marks 8)

(a) [5] In fibre optic communication system, explain signal transmission and conversion to sound.

(b) [3] A light is incident normally on a grating which has 2500 lines per centimetre. Compute the wavelength of a spectral line for which the deviation in second order is 15.0°.

GROUP – II

Paper: I Objective (iv)

فزکس

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6478

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جوابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. Which one is constant for a satellite in orbit:

(A) Potential energy
(B) Velocity
(C) Angular momentum
(D) Kinetic energy

Q2. The ratio of moment of inertia of a solid cylinder and thin ring is:

(A) 1/2
(B) 1/4
(C) 2/5
(D) 2/1

Q3. A fog droplet of radius 'r' falls in air with terminal velocity vt if the radius of the droplet becomes '2r'. Then terminal velocity is:

(A) vt/2
(B) 4vt
(C) 2vt
(D) vt

Q4. If the suspended mass of pendulum is doubled then its time period:

(A) Becomes double
(B) Becomes half
(C) Remains unchanged
(D) 1.414 times increased

Q5. The distance between two consecutive nodes or antinodes is:

(A) λ/2
(B) λ/4
(C) 2λ
(D) 4λ

Q6. When a wave is travelling in a denser medium incident on a rare medium, the reflected wave undergoes a phase change of:

(A) 0
(B) π/2
(C) π
(D) 2π

Q7. The transverse nature of light is confirmed by the phenomenon of:

(A) Interference
(B) Polarization
(C) Diffraction
(D) Beats

Q8. Final image formed in compound microscope is:

(A) Virtual and erect
(B) Virtual and inverted
(C) Real and inverted
(D) Real and erect

Q9. The mean square velocity of the gas molecule is given by:

(A) √(3RT/M)
(B) √(KT/M)
(C) √(2KT/m)
(D) 3KT/m

Q10. If a heat engine has 65% efficiency, then TL/TH =

(A) 0.5
(B) 0.45
(C) 0.35
(D) 0.25

Q11. 1 watt hour =

(A) 3.6 J
(B) 3.6 MJ
(C) $3.6x10^8$ J
(D) $3.6x10^3$ J

Q12. If the mass of the body is increased to four times while the force is kept constant, then acceleration in body will become:

(A) One Half
(B) Doubled
(C) One Fourth
(D) Four Times

Q13. The range of projectile at 60° is the same as that of angle:

(A) 30°
(B) 45°
(C) 90°
(D) 120°

Q14. Vectors A and B of magnitudes 4 N and 3 N make an angle of 30° and 60° with x-axis respectively. The scalar product of the two vectors:

(A) 12 N
(B) 7 N
(C) 10 N
(D) 6 N

Q15. Two forces act together on an object. The magnitude of their resultant force is minimum when they act at:

(A) 180°
(B) 90°
(C) 45°
(D) 0°

Q16. Which of the given quantity has different dimensions?

(A) Work
(B) Energy
(C) Pressure
(D) Torque

Q17. Time for 20 vibrations of a simple pendulum is 40.25 sec measured by a stop watch of accuracy up to 0.1 sec. The absolute uncertainty in the time period is:

(A) ±2.01
(B) ±0.05
(C) ±0.003
(D) ±0.005

Paper: I Subjective

فزکس

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any THREE questions from Section C.

نوٹ: حصہ دوم لازمی ہے۔ حصہ سوئم میں سے کوئی سے تین سوالات کے جوابات تحریر کیجیے۔

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Why do we find it useful to have two units for the amount of substance, the kilogram and the mole?

(ii) Name several repetitive phenomena occurring in nature which could serve as reasonable time standards.

(iii) How can you reduce random and systematic error?

(iv) Write down dimensions of a) work b) torque

(v) If A + B = 0, what can you say about the components of the two vectors?

(vi) Can a body rotate about its centre of gravity under the action of its weight?

(vii) What is the position vector? Explain.

(viii) Explain the circumstances in which the velocity v and acceleration a of a car are: a) v is zero but a is not zero. b) a is zero but v is not zero.

(ix) At what point or points in its path does a projectile have its minimum speed, its maximum speed.

(x) If we draw a graph between velocity and time, how can you find the distance and acceleration from it.

(xi) A boy uses a catapult to throw a stone which accidentally smashes a greenhouse window. List the possible energy changes.

(xii) What is salter's duck? Explain.

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) What is meant by moment of inertia? Explain its significance.

(ii) Show that orbital angular momentum, Lo=mvr

(iii) How many radians are there in 2 degree?

(iv) What is the difference between real and apparent weights?

(v) Explain, how the swing is produced in a fast moving cricket ball?

(vi) What are three conditions which fluid must satisfy to study its motion?

(vii) Can we realize an ideal simple pendulum?

(viii) Show that in SHM the acceleration is zero when the velocity is greatest and the velocity is zero when the acceleration is greatest.

(ix) If spring is cut into two equal halves, what will be spring constant of each part?

(x) What features do longitudinal waves have in common with transverse waves?

(xi) As a result of a distant explosion, an observer senses a ground tremor and then hears the explosion? Explain the time difference.

(xii) What is period of 300 cycles per second of sound waves?

Q4. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Can visible light produce interference fringes? Explain.

(ii) Why a fringe is shifted when mirror is shifted through λ/2 in Michelson interferometer.

(iii) Why longitudinal waves do not show polarization?

(iv) Why is it preferred to use the lens of small focal length for high angular magnification?

(v) One can buy a cheap microscope for the use of children. The image seen in such a microscope have coloured edges. Why is this so?

(vi) What do you understand by axial ray? How does it travel?

(vii) Can Boyle's law be derived by Kinetic Molecular theory? If so, explain.

(viii) Specific heat of a gas at constant pressure is greater than specific heat at constant volume. Why?

(ix) Why is it not possible to utilize the heat contents of oceans and atmosphere?

> Note: Attempt any THREE questions. Each question carries Eight (8) marks. (8x3=24)

Q5. (Marks 8)

(a) [5] What is meant by rectangular components? Explain addition of two vectors by rectangular components.

(b) [3] Prove that for angles of projection, which exceed or fall short of 45° by equal amounts, the ranges are equal.

Q6. (Marks 8)

(a) [5] If a body of mass 'm' is dropped from height 'h' as shown, discuss the interconversion of energies during its motion.

(b) [3] An organ pipe has a length of 50 cm. Find the frequency of its fundamental note and the next harmonic when it is closed at one end.

Q7. (Marks 8)

(a) [5] What is meant by centripetal force? Derive relations for centripetal force and centripetal acceleration.

(b) [3] A block of mass 4.0 kg is dropped from a height of 0.80 m onto a spring of spring constant K=1960 Nm^{-1}, find the maximum distance through which the spring will be compressed.

Q8. (Marks 8)

(a) [5] Define molar heat capacity and prove that Cp-Cv=R

(b) [3] How large must a heating duct be if air moving 3.0 ms^{-1} along it can replenish the air in a room of 300 m^3 volume every 15 min? Assume the air's density remains constant.

Q9. (Marks 8)

(a) [5] How would you derive the equation of magnification of compound microscope accompanied by ray diagrammatic explanation? Also, give two points to resolve the details of an image.

(b) [3] X-rays of wavelength 0.150 nm are observed to undergo a first order reflection at a Bragg angle of 13.3° from a quartz crystal. What is the interplanar spacing of the reflecting planes in the crystal?